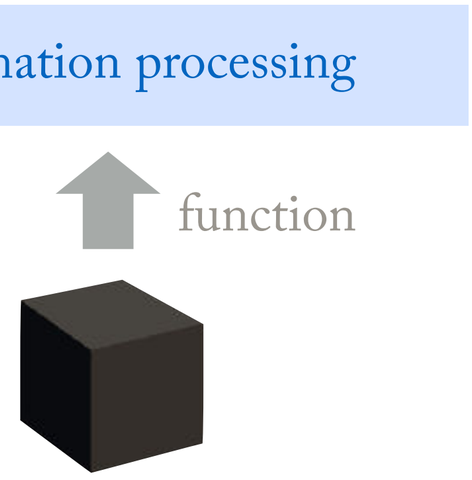
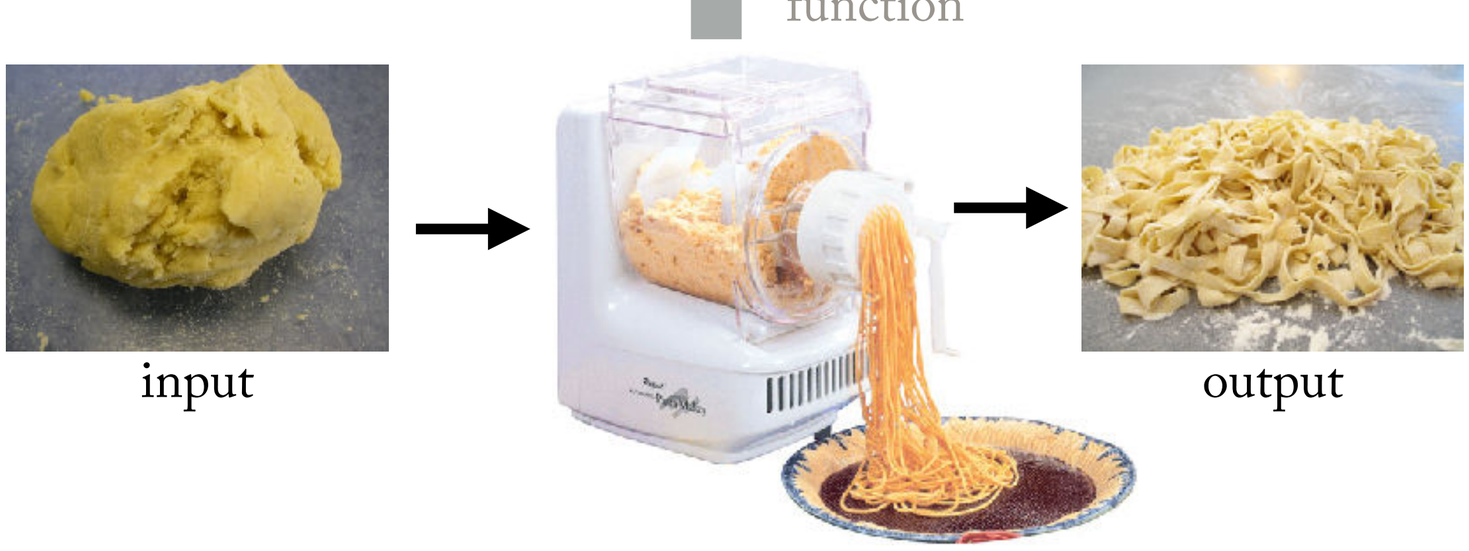
2. What is a Computer?

Information Processing

A computer is a machine that has the function of **processing information**.

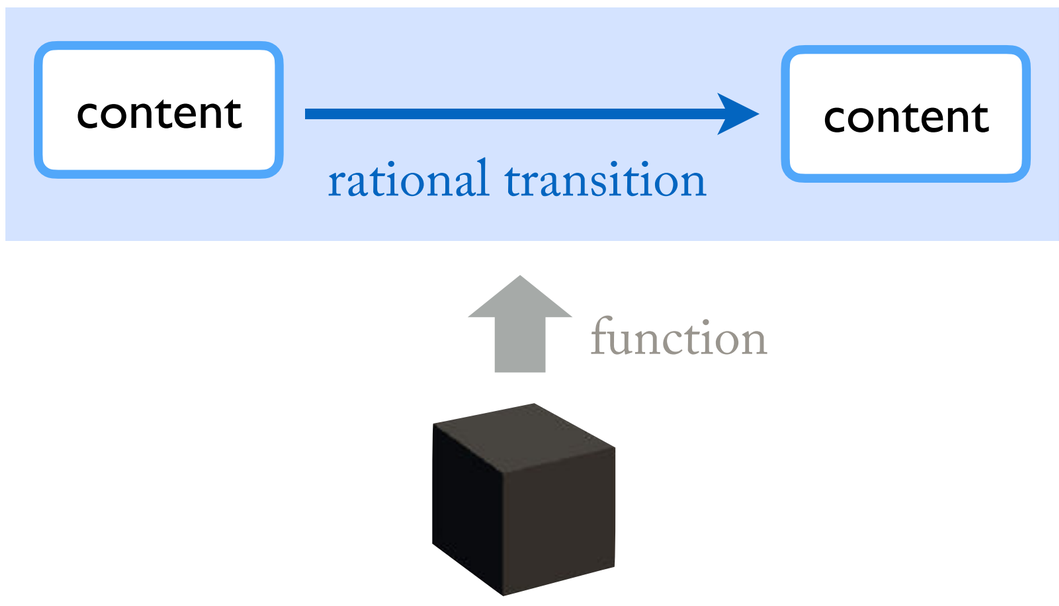


Like any other machine, it has a definite function…



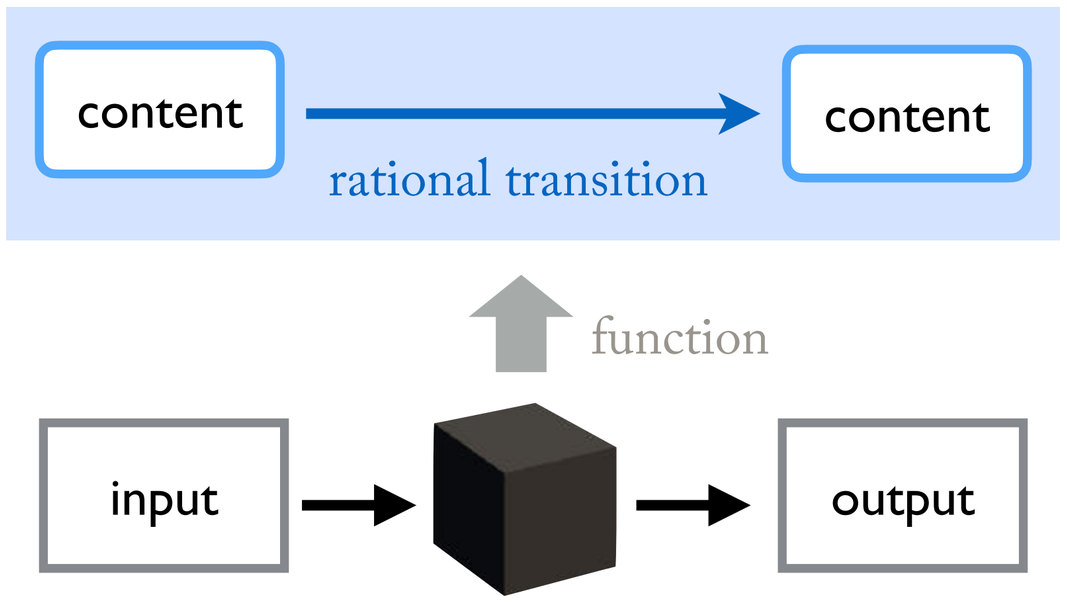
… but the function of a computer isn't something that can be described in purely physical terms.

To process information is to perform **rational transitions** between **contentful states.**



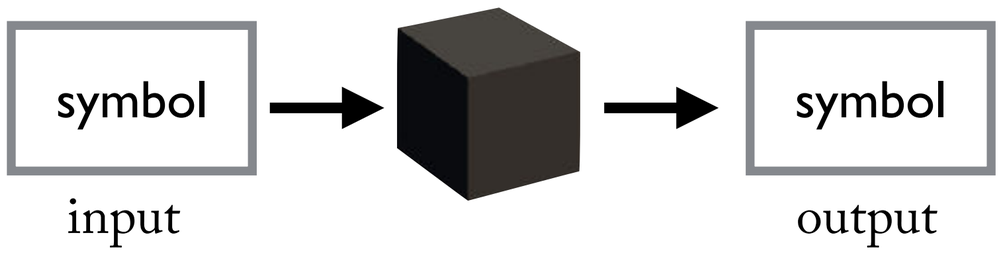
Ultimately, a computer is still a **physical object**, and it must perform its informational function *by* performing some physical process.

In fact, like the pasta machine, all computers are **input/output machines**. They take physical inputs and process them into physical outputs.



Symbol Processing

A computer takes physical signals as **input** and produces physical signals as **output**.



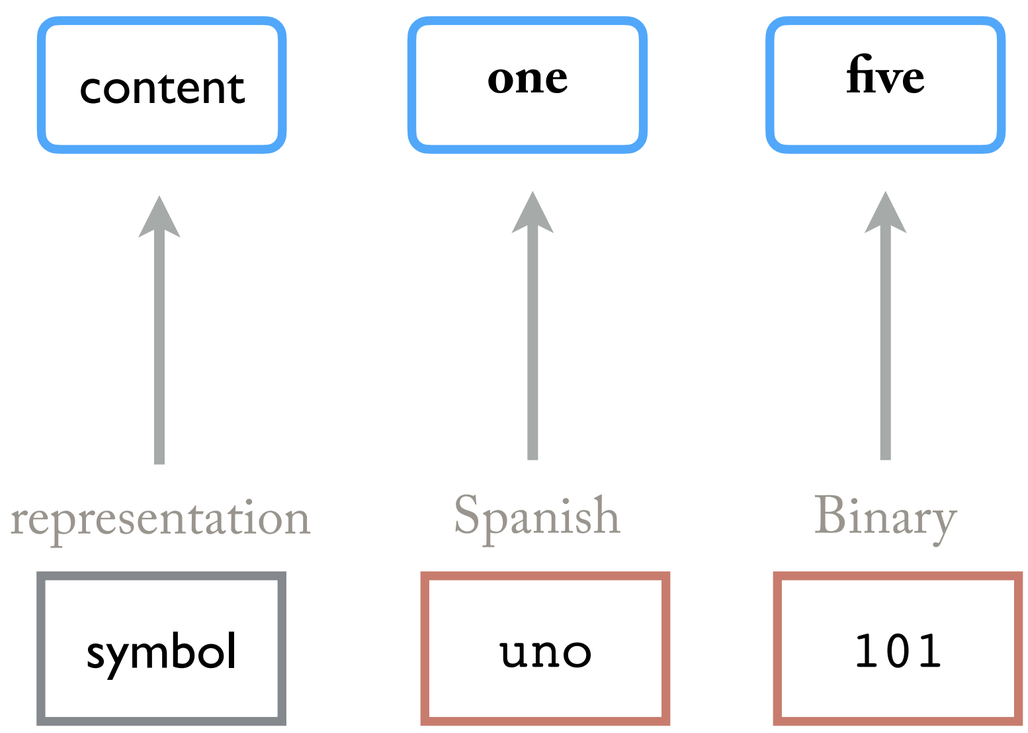
Symbols are physical objects or patterns. A computer takes symbols as inputs, processes symbols, and produces symbols as outputs.



A **representation** is a physical symbol that is *about* something or expresses a meaning.

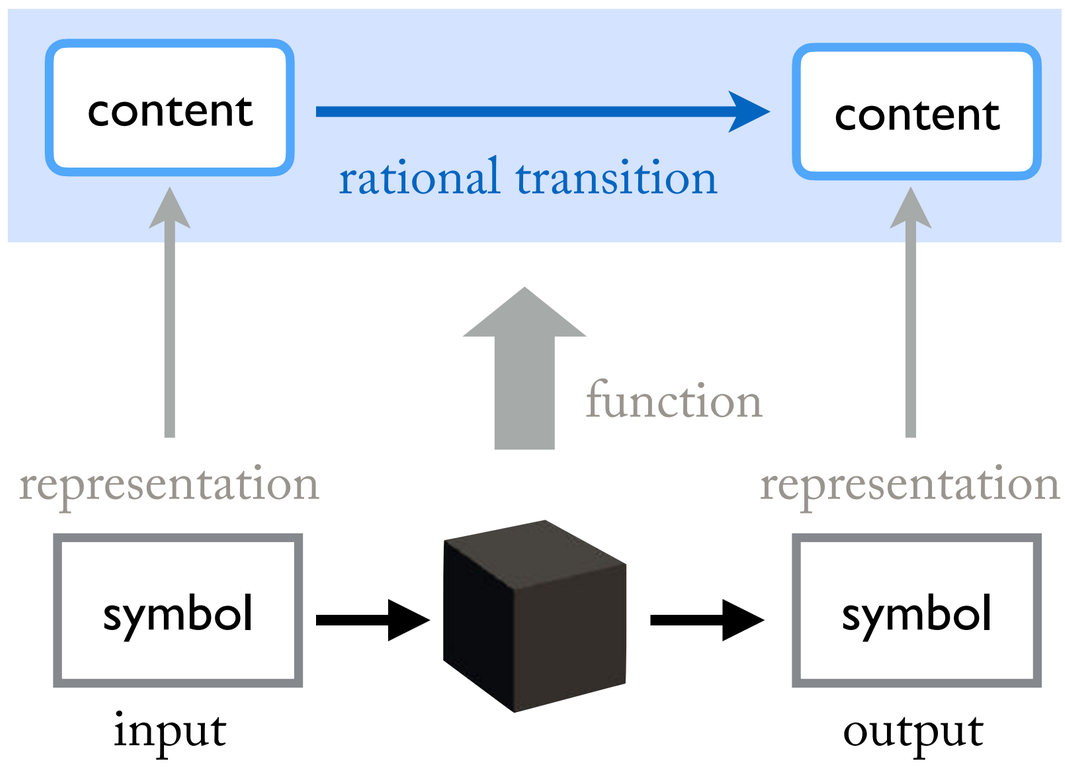
What a representation is about or what it means is its **content**. Symbols **represent** their contents.

Both **external** symbols, like words or pictures, and **internal** symbols, like computer code, can express content.

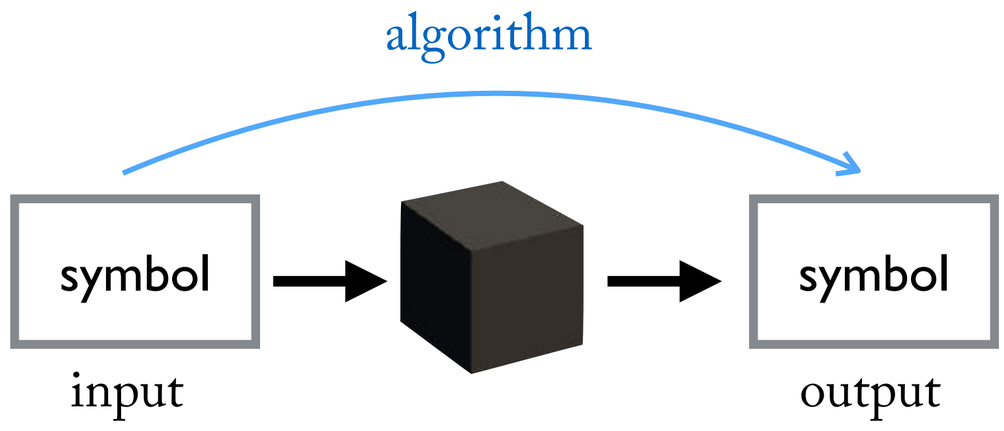


For a computer to perform a rational transition, its inputs and outputs must represent the initial and final contents of the rational transition.

The representations processed by the computer must **mirror** the rational transition.



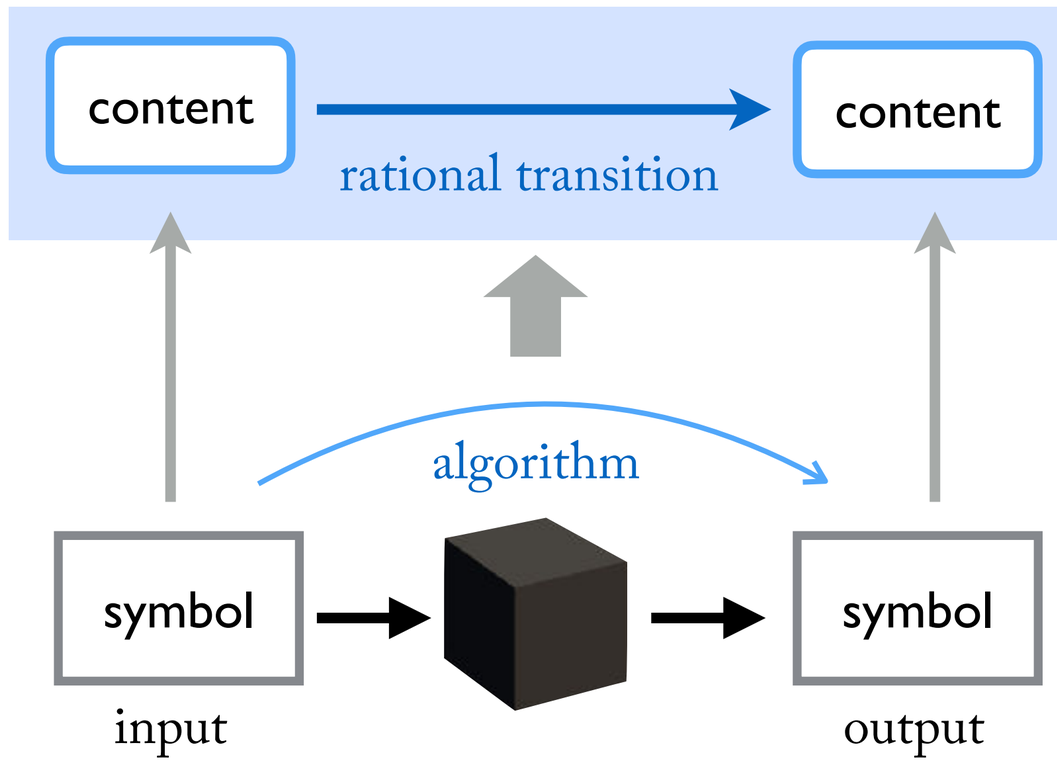
In order to produce representations which mirror rational transitions, computers follow **algorithms:** mechanical procedures for manipulating symbols.



Computation

A computer processes information *by* processing symbols.

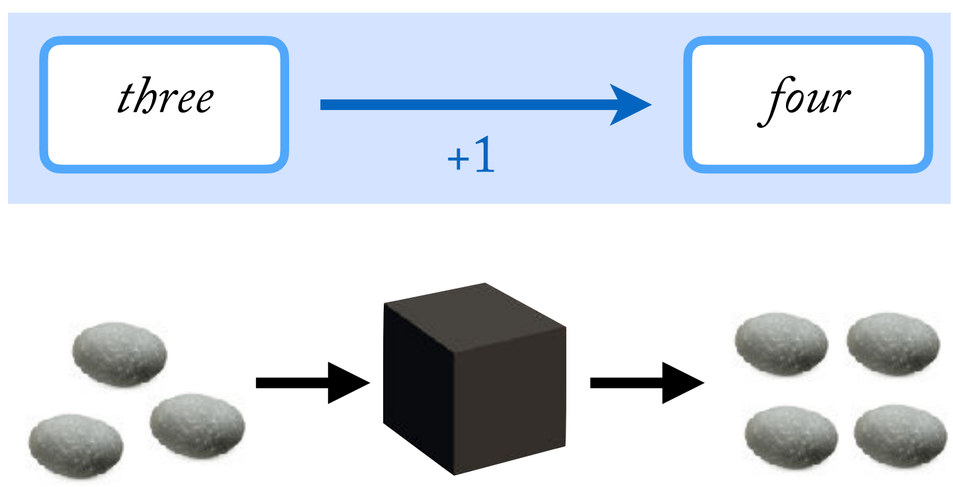
A computer functions to perform rational transitions by manipulating representational symbols according to an algorithm.



Computation as embodied logic.

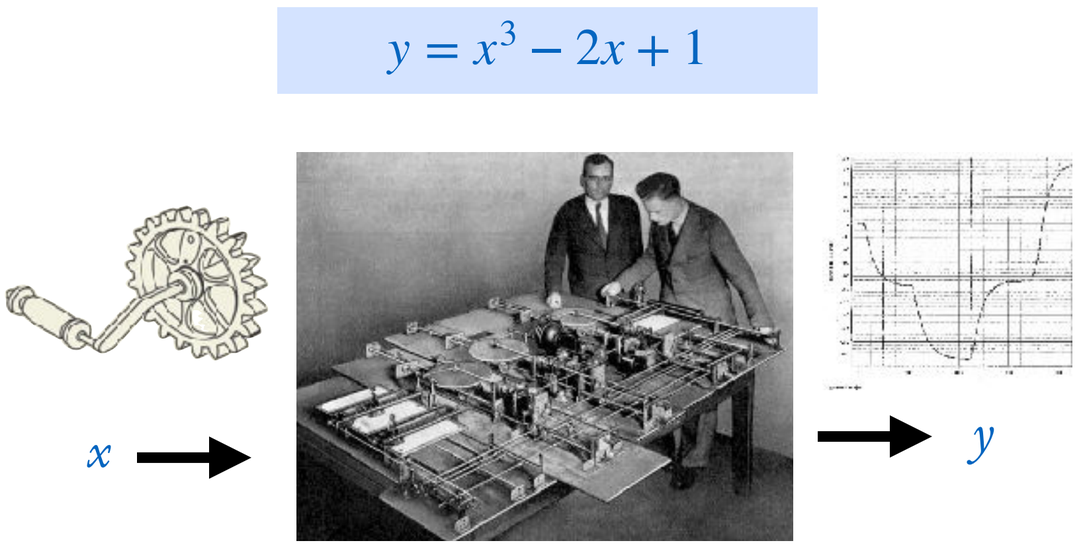
Example: Caveman computer

This computer takes buckets of rocks as input, and produces buckets of rocks as output. Each collection of rocks represents a number. The computer performs the operation +1.



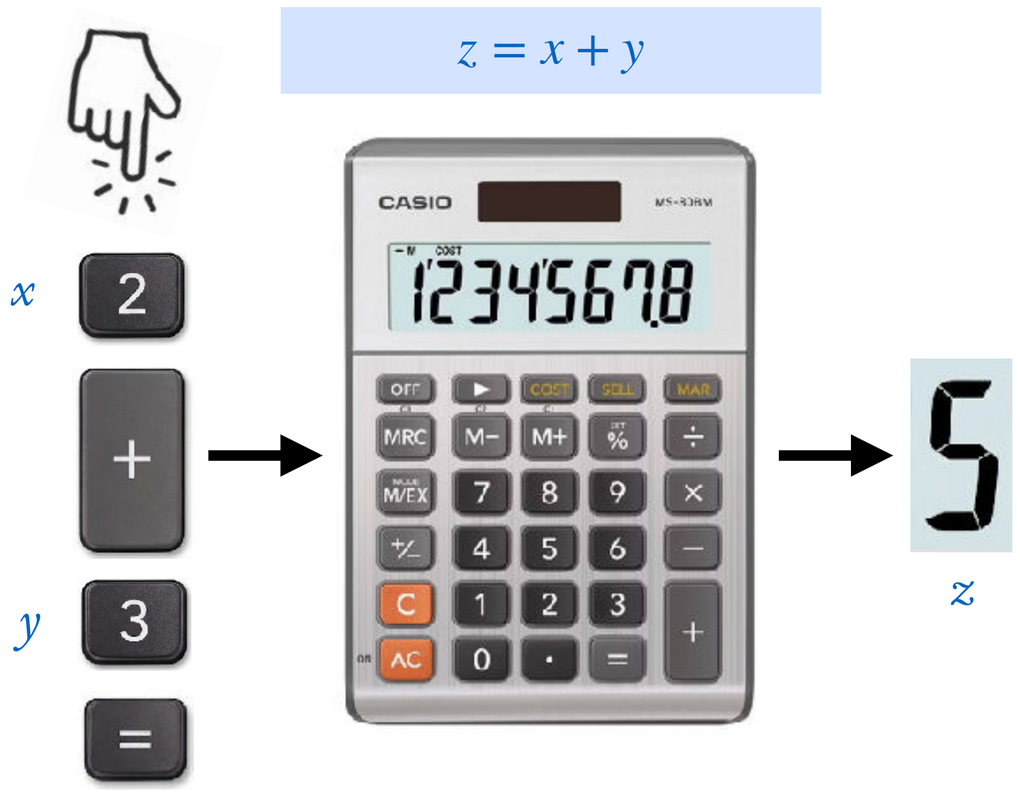
Example: Analog computer

Historical analog computers involved complex networks of gears. The input is entered as the rotation of a gear wheel. The output is the rotation of yet another gear wheel. They could solve polynomial equations.



Example: Digital calculator

The inputs to a digital calculator are button presses. The outputs are displayed numerals on a screen. The inner circuitry of a calculator is very much like the machines we will design in Unit 1.



Computational Theory of Mind (CTM)

Our definition of computation is broad enough that it may potentially include both artificial and naturally evolved information processors. In fact, according to the **computational theory of mind**, the brain itself is a kind of computer.

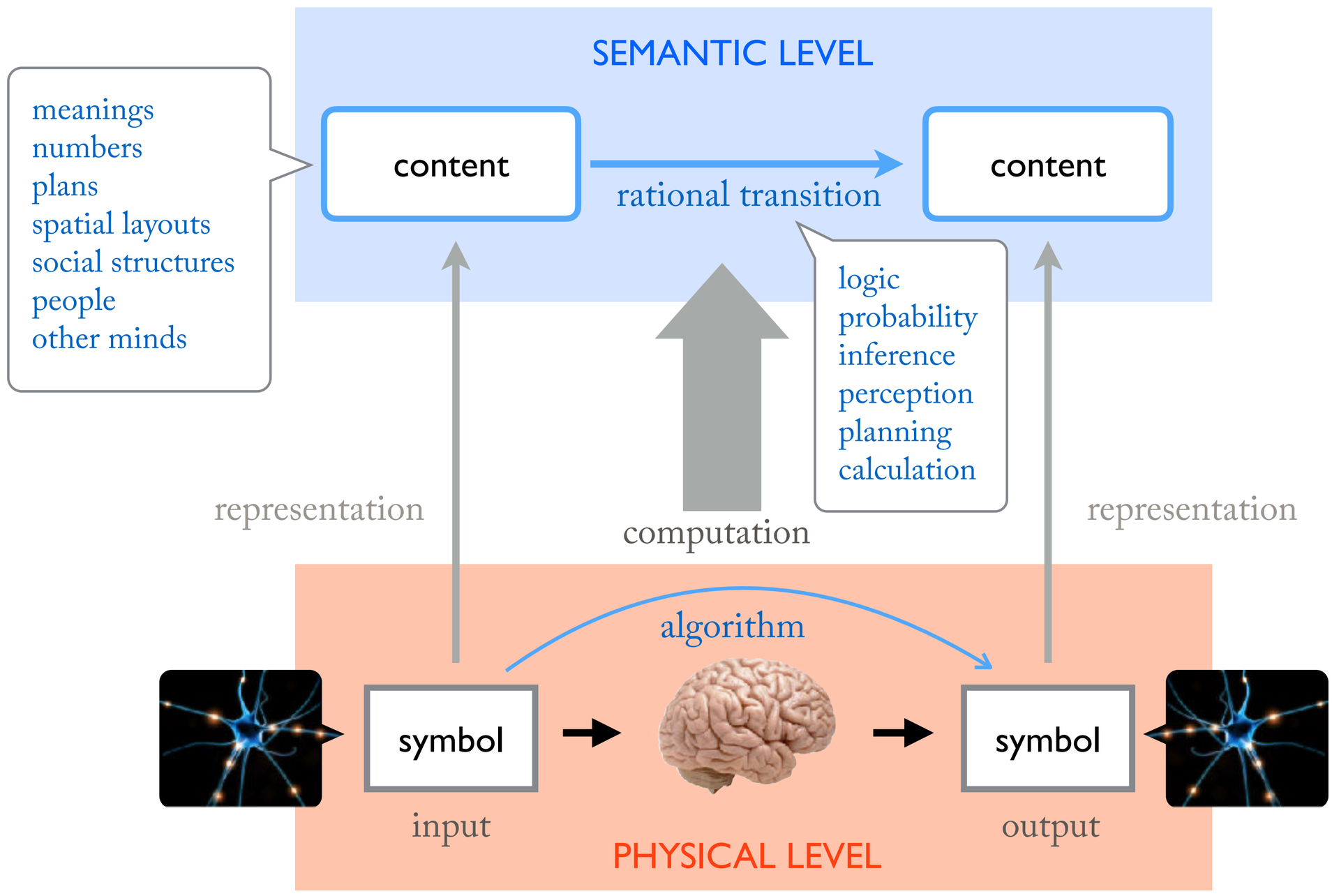
Information processing
The brain has the function of processing information through rational transitions. This is a biological function, not a design function.

Symbol processing
The brain takes neural symbols as inputs, processes neural symbols, and produces neural symbols as outputs. Neural symbols in the brain carry mental contents.

Computational Theory of Mind (v. 1)

1. The brain is a naturally occurring computer or information processor.
2. Cognition arises from the computational activity of the brain.

CTM uses the same computational framework that was traditionally applied only to artificial computers and extends it to the brain. CTM is the foundational hypothesis of contemporary cognitive science.



Computational Theory of Mind: the brain computes cognitive functions by manipulating symbolic representations in the brain according to biologically grounded algorithms.